FOOTBALL ANALYSIS TOOL · WYSCOUT DATA

SCOUTANALYTICS

METHODOLOGY

This document explains how the data is checked, how teams and players are analysed, how profile search works and what has been verified. It describes version 2 of the tool, as presented on the portfolio.



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01 PRINCIPLES

ScoutAnalytics is used to analyse a team (your own or an opponent), to analyse a player and to find players for a specific need. Four rules guide the whole method.

- **The tool does not decide what a good player is.** No overall rating is given. A player is described statistic by statistic, and the search ranks players by the priorities the user sets.
- **A player is compared with players in the same role.** A full-back is compared with full-backs, a defensive midfielder with defensive midfielders.
- **Context is only used where it has been shown to matter.** A contextual reading exists only for statistics where a prior study showed that the team's style and level have a real effect (section 4).
- **A missing value stays missing.** It is never replaced by zero or silently estimated. A rate with no attempts is left blank; a reading that cannot be computed is flagged with the reason.

02 CONTROLLED DATA

Wyscout exports go through a controlled import before entering a SQL database. The analysis notebook then reads that database without ever changing it.

THE EXPORTS

- **Players**: one file per league and season (115 columns), one row per player and club.
- **Team Stats with opponents**: one file per team and season (109 columns), with the team's row and its opponent's row for every match.

THE IMPORT

The import runs with a single command, folder by folder (one season, one league). The database is backed up before anything is written. Each file is then recognised, checked and imported, and the result is recorded in a log.

- **Format checks**: file type, expected columns, season, declared league. A file that fails is rejected with the reason; nothing from it is written.
- **No duplicates**: a file already imported is recognised by its fingerprint and skipped. Running an import again changes nothing.
- **In-season updates**: a new export explicitly replaces the previous one; a value corrected by Wyscout replaces the old one, which is archived.
- **Matches**: every match must have two rows and two consistent sides. A probable duplicate is never merged automatically; it is flagged and decided by hand. Two matches played on the same day under the same label (friendlies) remain two separate matches.
- **Undo**: the last import can be undone by restoring the backup.

PLAYER IDENTITY

Wyscout does not provide a player ID and does not always spell a name the same way. A player is linked from one season to the next only when everything matches: initial and surname, country of birth and a shared passport, consistent ages, and at least one more clue (same spelling, same height or same club). Two players with the same name at the same club stay two players. Doubtful cases are decided by hand, and every decision is kept.

Wyscout sometimes has two records for the same player in the same export (overlapping matches and minutes). The import flags it; the discarded record stays in the database but is no longer read. A mid-season transfer, on the other hand, gives two correct rows, one per club.

THE DATABASE AND HOW IT IS READ

The SQLite database keeps the data as Wyscout provides it, along with the decisions made during import. Ready-made views give one row per player and season, per team and match (with the opponent's statistics) and per team and season. The analysis notebook (Python, Google Colab) opens the database read-only, shows its fingerprint on every run, and refuses a missing database or a missing season.

03 TEAM ANALYSIS

A team is described in two ways: its playing identity (how it plays) and its collective performance (what it produces and concedes). All values come from Team Stats, match by match.

PLAYING IDENTITY: 7 AXES

Each axis is a season proportion or average. Proportions are computed from season totals, not by averaging match percentages.

AXIS	CALCULATION
Short build-up	100 − average share of long passes
Crosses	box entries from crosses ÷ box entries
Transition	counter-attacks ÷ (counter-attacks + positional attacks)
Pressing	average PPDA, reversed (fewer opponent passes allowed = more intense pressing)
High recoveries	recoveries in the final third ÷ recoveries
Own-third losses	losses in the defensive third ÷ losses
Set pieces	set-piece shots ÷ shots (positional attacks, counter-attacks, set pieces)

ROBUST GAP, TREND OR AVERAGE

Each team is compared with the median of its league, or of all loaded leagues. To tell whether a gap is solid or down to a handful of matches, the team's matches are resampled 500 times (drawn with replacement, with a fixed seed so results can be reproduced), and the gap to the median is recalculated on each draw.

- **Robust gap**: the 95% interval of the gap does not contain zero.
- **Trend**: only the 80% interval does not contain zero.
- **Average**: the gap cannot be told apart from zero. This does not mean the team is exactly average.

Two axes are less stable than the others from one half of the season to the other: transition (0.51) and set pieces (0.52), reliability measured by splitting the matches into two halves (Spearman-Brown correction). Their conclusions call for more caution.

The main system (the one used most over the season) comes from the formation Wyscout assigns to each match. The diagram shows that system, not the players' actual positions.

COLLECTIVE PERFORMANCE

64 per-match statistics in 8 groups, each with the direction in which a value is favourable (many shots, few shots conceded). They are used to rank teams on one statistic, to compare two teams and to place a team against the median.

04 PLAYER ANALYSIS

Each of a player's statistics is given as a percentile among players in the same role. Where the study justified it, a second reading takes the team's context into account.

REFERENCE ROLES

The position comes from the player's Wyscout positions and is then refined: goalkeeper (GK), centre-back (CB), full-back (FB), wing-back (WB), defensive midfielder (DM), central midfielder (CM), attacking midfielder (AM), winger (W), centre-forward (CF). Left and right full-backs are grouped together, as are wingers. An adjustable minutes threshold leaves out players who have played too little.

THE OVERALL READING

The overall percentile places the player among all players in the same role across the loaded leagues. Ties get the average rank; missing values are ignored, not counted as zeros. The catalogue has 78 statistics, two of them computed by the tool: the share of progressive passes and non-penalty xG.

THE CONTEXTUAL READING

A midfielder touches the ball more in a team that has the ball; a defender defends more in a team that is dominated. The contextual reading removes that effect: it compares the player with what is expected of a player in the same role in a team like his.

```
expected value = linear model (5 team factors + league)
contextual percentile = rank of (actual value − expected value) among players in the role
```

TEAM FACTOR	MEASURE
Possession	average possession
Block height	recoveries in the final third ÷ recoveries
Direct play	share of long passes
Transitions	counter-attacks per match
Level	average xG difference per match (goal difference if opponent data is missing)

The model is fitted separately for each statistic and each role, with at least 30 players and 15 teams. If any of the five factors is missing for the player's team, there is no contextual reading, and the reason is shown.

THE CONTEXTUALISATION STUDY

Before the tool was changed, a study tested, for each statistic and each role, whether team context really matters. It covers the 2025-26 season of 8 European leagues: 2,556 players with 900 minutes or more, including 186 goalkeepers, and 148 teams. 547 statistic × role combinations were examined.

For each one, the study looked at how much of the difference between players is linked to the team, the size of the effect, its stability league by league, its robustness across five model variants, and whether its direction

matched the hypothesis written before the analysis. The protocol did not set numerical thresholds in advance; decisions followed a consistent line, recorded with each case.

STATUS	COMBINATIONS IN THE TOOL	
Clear	135	contextual reading
Moderate	79	contextual reading, flagged "moderate influence"
Exploratory	7	contextual reading, flagged "exploratory"
Exploratory, suspended	7	overall percentile only (hypothesis to be reworded)
Not retained	277	overall percentile only
Not assessable	42	overall percentile only (more than half of players at zero)

Context matters for almost all passing, progression, chance creation by midfielders, wingers and forwards, attacking volume (shots, xG, box touches), part of defensive volume, and everything the goalkeeper faces.

No correction is retained for duel and dribble success, finishing quality, crosses, set pieces and goals prevented. This does not prove there is no link: the correction was not sufficiently justified.

The tool reads the decisions from a single reference file; nothing is hard-coded. Every revision is dated and justified there.

RATES AND THEIR DENOMINATORS

- For each rate, the tool shows the number of attempts over the season: the actual count when the export provides it, an estimate otherwise (per-90 value × minutes ÷ 90), flagged as such.
- Four per-90 values are recalculated from season totals, which are more precise than the export's rounded values: xG, xA, non-penalty goals and shots.
- Non-penalty xG is estimated by removing 0.76 xG per penalty taken.
- The denominator of "offensive duels won, %" was checked: the duels won it implies come out as a whole number for 100% of the 2,698 players tested (900 minutes or more), against 15% and 22% with two other possible denominators.

05 PROFILES, SIMILAR PLAYERS AND SEARCH

Three uses rely on the same percentiles: reading a player's profile, finding similar profiles and searching for a specific need.

THE RADAR AND STRENGTHS

Each role has a default radar of 7 to 11 statistics (listed in the appendix), which can be changed using the role's catalogue. The radar can be shown in the overall or the contextual reading. Strengths and weaknesses are taken from the radar axes.

SIMILAR PLAYERS

Similar players are searched for in the same role, using overall-reading percentiles on the radar's default axes. Only axes with data for both players are compared, and a candidate must share at least 60% of them (and at least 3).

```
similarity = 100 − √(mean of the squared percentile gaps)
```

PROFILE SEARCH

The user chooses a role, the statistics that matter to them and their weight. Results can be filtered by age, minutes, market value and contract expiry.

```
search score = weighted mean of the percentiles of the chosen statistics
```

With no weighted statistic, there is no score. The result depends on the priorities chosen: for the same position, two clubs with different needs get two different rankings.

06 VERIFICATION

Every change is tested before it is adopted, by comparing results before and after.

- **Database and original files.** For the 2025-26 season of the 8 leagues, the dashboard produced from the SQL database is byte-for-byte identical to the one produced directly from the Excel exports, at 45 and at 900 minutes. The database returns all 148 Team Stats exports exactly as they are in the files (values, types, order).
- **Import.** Automated test suites cover rejected files, re-imports, updates, duplicate matches, same-day matches and links between players.
- **Notebook.** The safeguards are tested: missing database, missing season, wrong file; and any attempt to write fails.
- **Dashboard.** During the code clean-up, data and display were compared before and after, in all three languages and for several players and teams. A negative control confirmed that the comparison detects a single changed value.
- **Calculations.** The playing-identity intervals are checked on every run (the 80% interval always lies within the 95% one). Save percentage was recalculated from shots against and goals conceded for all 186 goalkeepers. The study's decisions were compared automatically with the file the tool reads: no differences.

07 LIMITS

- **Match-event data only.** Wyscout does not provide physical data. The tool shows whether a player stands out in his role, in a comparable context; it does not say whether he will adapt to a given system.
- **Validation still to come.** The study's decisions must still be confirmed on an independent dataset, with a decision rule fixed before looking at that data. Exploratory effects and small groups (attacking midfielders, goalkeepers) come first.
- **Context is not cause.** The contextual reading removes the average effect of the team's style and level; it does not measure a player's influence on his team.
- **Estimated denominators.** For most rates, the number of attempts is estimated; consistency is checked, not Wyscout's exact denominator.
- **Data as of the export date.** Age, market value, contract and current club are those on the day of the export, not of the season. Wyscout sometimes recalculates its models (xG): two seasons exported far apart are not necessarily calculated the same way.

Wyscout data is licensed: it is neither published nor reproduced in this document. The full tool is shown in interviews.

A APPENDIX: DEFAULT RADARS

Statistics shown by default on each role's radar. The user can remove them or add others from the role's catalogue.

ROLE	DEFAULT AXES
GK	Goals prevented/90 · Save % · Claims & sweeps/90 · Aerial duels/90 · Passes/90 · Pass accuracy % · Long passes/90 · % long pass accuracy
CB	Def. duels/90 · % def. duels won · Interceptions/90 · Shots blocked/90 · Aerial duels/90 · % aerial duels won · Passes/90 · Pass accuracy % · Prog. passes/90 · % long pass accuracy · Prog. carries/90
FB	Def. duels/90 · % def. duels won · Interceptions/90 · Prog. passes/90 · Prog. carries/90 · Crosses/90 · xA/90
WB	Def. duels/90 · % def. duels won · Interceptions/90 · Prog. passes/90 · Prog. carries/90 · Crosses/90 · xA/90 · Box touches/90
DM	Def. duels/90 · % def. duels won · Interceptions/90 · % aerial duels won · Passes/90 · Pass accuracy % · Prog. passes/90 · % long pass accuracy · Prog. carries/90
CM	Def. duels/90 · % def. duels won · Passes/90 · Prog. passes/90 · Prog. carries/90 · xA/90 · npxG/90
AM	xA/90 · Key passes/90 · Passes into box/90 · npxG/90 · Box touches/90 · Dribbles/90 · % successful dribbles · Prog. carries/90
W	Dribbles/90 · % successful dribbles · Prog. carries/90 · xA/90 · Passes into box/90 · npxG/90 · Box touches/90
CF	npxG/90 · Shots/90 · Shots on target % · Box touches/90 · Key passes/90 · Att. duels/90 · Prog. carries/90